Fiscal Year:	FY 2008	**Task Last Updated:**	FY 05/06/2008
PI Name:	Moorman, Stephen J Ph.D.		
Project Title:	The Effects Of Microgravity On Gene Expression In Live Zebrafish Embryos		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
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Zip Code:	08854-5635	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	11/01/2002	**End Date:**	10/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	6
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 10/31/2007 per A. Chu (10/06)		
Key Personnel Changes/Previous PI:	1		
COI Name (Institution):			
Grant/Contract No.:	NAG2-1591		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The effects of microgravity on gene expression are poorly understood. Initial attempts have been made to investigate the effects of microgravity on cells in culture. It is unknown how those experiments will relate to potential effects on gene expression in vivo. We have begun to examine the effect of simulated-microgravity on gene expression in specific developing organ systems in vivo using transgenic zebrafish that express the green fluorescent protein gene (gfp) under the influence of different promoters. The use of gfp as a "reporter-gene" has two significant advantages; you can monitor gene expression directly in a live animal, and you can easily detect changes in morphology that might be due to changes in expression of other genes. This is the first time a whole animal in vivo model will be used and maintained alive during gene expression analysis for microgravity related research. This research not only addresses an important issue in space life sciences research, it clearly demonstrates the usefulness of zebrafish with gfp constructs as a model to study the effects of microgravity on gene expression during development. Use transgenic zebrafish that express the gene for GFP under the influence of specific promoters to determine the effects of simulated-microgravity on gene expression in live zebrafish embryos. We have documented specific changes in gene expression that are related to changes in gravitational force. This suggests that gravitational force plays a fundamental role in regulating gene expression and might have been a driving force for vertebrate evolution. Interestingly, organs that show effects in our experiments either develop into or regulate the development of organ systems that are affected in astronauts during space flight. This suggests that ground-based research using zebrafish embryos might have predictive value for the affects of long-duration space flight on astronaut health.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Earth's gravitational field has remained constant during all of evolutionary history. Therefore, it is possible that gravity might play some fundamental role in regulating gene expression during vertebrate development. Interestingly, organs that show effects on gene expression in our experiments either develop into or regulate the development of organ systems that are affected in astronauts during space flight. This suggests that ground-based research using zebrafish embryos might have predictive value for the affects of long-duration space flight on astronaut health.
Task Progress:	The Specific Aims outlined in the grant proposal have all been accomplished. We have identified specific periods during zebrafish development when organ systems are susceptible to microgravity induced changes in gene expression. In addition we have identified a cell organelle, the primary cilium, that transduces changes in gravitational force into changes in gene expression. We have also demonstrated that the primary cilium can transduce the normal cyclic changes in earth's gravitational force due to the moon (the tidal force cycle) into changes in molecular pathways that regulate the stochastic nature of gene expression - effectively regulating transcriptional noise.
Bibliography Type:	Description: (Last Updated: 07/05/2019)
Abstracts for Journals and Proceedings	Moorman SJ. "Simulated-microgravity causes changes in heat shock protein expression in zebrafish embryos." ASGSB annual meeting, November 2003. Gravitational and Space Biology Bulletin 2003 Nov;17(1):57. , Nov-2003
Abstracts for Journals and Proceedings	Moorman SJ, Hoff JK, Sokunbi G. "Changes in gravitational force induce changes in gene expression in the developing zebrafish heart." Experimental Biology 2004 meeting, May 2004. FASEB J 2004 Supplement. , May-2004
Abstracts for Journals and Proceedings	Shimada N, Moorman SJ. "Effects of microgravity on hsp-70 gene expression in developing lens." Experimental Biology 2005 meeting, May 2005. FASEB J. 2005;Suppl:491.4. , May-2005
Abstracts for Journals and Proceedings	Shimada N, Moorman SJ. "Effects of altered-gravity on hsp70 gene expression in developing lens." ASGSB annual meeting, November 2005. Gravitational and Space Biology 2005 Nov;19(1):62. , Nov-2005
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Abstracts for Journals and Proceedings	Shorr AZ, Moorman SJ. "The primary cilium is a gravity sensor." Experimental Biology 2008 meeting, May 2008. FASEB J 2008;Suppl: 481.2. , May-2008
Articles in Peer-reviewed Journals	Shimada N, Sokunbi G, Moorman SJ. "Changes in gravitational force affect gene expression in developing organ systems at different developmental times." BMC Developmental Biology 2005 May 31;5:10. PMID: 15927051 , May-2005
Articles in Peer-reviewed Journals	Shimada N, Moorman SJ. "Changes in gravitational force cause changes in gene expression in the lens of developing zebrafish." Developmental Dynamics 2006 Oct;235(10):2686-94. PMID: 16894605 , Oct-2006
Articles in Peer-reviewed Journals	Moorman SJ, Shimada N, Sokunbi G, Pfirrmann C. "Simulated–microgravity induced changes in gene expression in zebrafish embryos suggest that the primary cilium is involved in gravity transduction." Gravitational and Space Biology 2007 Jun;20(2):79-86. , Jun-2007
Articles in Peer-reviewed Journals	Moorman SJ, Shorr AZ. "The primary cilium as a gravitational force transducer and a regulator of transcriptional noise." Developmental Dynamics. 2008 Aug;237(8):1955-9. Review. https://doi.org/10.1002/dvdy.21493 ; PubMed PMID: 18366139 , Aug-2008
Significant Media Coverage	Kiefer J. "May the (gravitational) force be with you.“ Summary of article, “Changes in gravitational force cause changes in gene expression in the lens of developing zebrafish.“ in “Highlights of DD" (Developmental Dynamics journal)." Developmental Dynamics 2006 Oct;235(10):Highlights of DD., Oct-2006